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Agentic AI-Based Dynamic Tariff Optimization for EV Charging Networks

Using Large-Scale Real-World Charging Session Data

30K+	R^2=0.99	+8.99%
Charging Sessions	Prediction Accuracy	Off-Peak Uplift

The Problem & Our Solution at a Glance

⚠️ PROBLEM

Fixed ₹/kWh pricing ignores real-time congestion

Peak-hour queues & off-peak underutilization

No feedback mechanism to improve decisions

🤖 OUR SOLUTION

Agentic AI pipeline with 3 autonomous agents

Surge pricing up to 1.5× + off-peak −30% discount

Self-improving Monitoring & Learning Agent

📈 IMPACT

+8.99% off-peak energy uplift

R^2 = 0.9947 demand prediction accuracy

PES improved 12.14 → 12.65 over 6 episodes

Built on 30,000+ ACN charging sessions (Caltech/JPL) + 24,798 Shenzhen charging piles (ST-EVCDP)

Two Real-World Datasets, One Intelligence Layer

🔌 ACN-Data (Caltech / JPL)

30,000+ charging sessions | JSON → CSV
Features: timestamps, energy, duration, station IDs, user behaviour

🌆 UrbanEV — ST-EVCDP (Shenzhen)

24,798 charging piles | 5-min interval resolution
Large-scale urban: spatial + temporal demand variation

ENGINEERED FEATURES

- **Charger Utilization Rate**
  Charging Time ÷ Total Available Time
- **Revenue per Session (₹)**
  kWh delivered × dynamic tariff
- **Energy Cost per kWh**
  ToU baseline: ₹12/kWh peak, ₹8/kWh off-peak
- **Queue Length Proxy**
  max(0, occupancy − pile_count)
- **Occupancy Density**
  occupancy ÷ grid area

Demand Prediction Agent: Forecasting Ahead of Demand

Model	RMSE	MAE	R^2 Score
✅ Random Forest	0.0121	0.0046	**0.9953**
XGBoost	0.0128	0.0053	0.9947
LightGBM	0.0127	0.0051	0.9949
HistGBM	0.0131	0.0053	0.9945

AGENT OUTPUTS

📊 **Predicted utilization rate**

Per grid × time slot

🚨 **Binary congestion flag**

Threshold: utilization > 80%

⚡ **Expected charging load**

kWh per time slot

All models trained on a temporal hold-out split — no look-ahead leakage

Dynamic Tariff Optimization: Smarter Real-Time Pricing

🔴 SURGE

Utilization > 80%

Up to 1.5× base rate

🟡 STANDARD

30% – 80%

₹15/kWh fixed baseline

🟢 DISCOUNT

Utilization < 30%

−30% off (0.7× multiplier)

EPISODE 1 RESULTS (13 Jul 2022 · All Shenzhen Grids)

+8.99%

Off-Peak Energy Uplift

30% discount successfully shifted demand

−12.6%

Short-term Revenue Δ

Expected: discount strategy, recovers long-run

1.5×

Max Surge Multiplier

Applied at utilization > 80% congestion zones

Monitoring & Learning Agent: Self-Improving Over Time

Episode	Date	Surge Threshold	PES	Off-Peak Uplift
1	13 Jul	0.80	12.14	+8.99%
2	14 Jul	0.78	12.24	+8.99%
3	15 Jul	0.76	12.38	+9.00%
4	16 Jul	0.74	12.56	+9.00%
5	17 Jul	0.72	12.50	+9.00%
6	18 Jul	0.70	12.65	+9.00%

EVALUATION METRICS

PES +4.2%

12.14 → 12.65 over 6 episodes
Revenue per kWh delivered

CRR −30%

Consistent session shift
in response to discount signal

Wait Time ↓

Queue proxy reduced
across peak windows

The agent autonomously reduced surge threshold 80% → 70% over 6 days — no human intervention required

From Algorithm to Real-World Impact

🏢 Network Operators

- Off-peak uplift ~+9%: more sessions per charger per day
- PES improvement → self-optimizing revenue management
- No new infrastructure required

🏙️ Grid & Urban Planners

- Smoother demand curves reduce procurement cost spikes
- Spatial data identifies highest-ROI charger locations
- ToU alignment reduces peak grid stress

🏛️ Policy Makers

- Rule-based pricing (>80% surge, <30% discount) is auditable
- Fully documented assumptions — no black-box claims
- Replicable for Indian EV markets via CESL / TATA Power

⚠️ Honest Caveats

Revenue Gain % is currently negative — the system optimises for load balancing, not short-term maximisation. A 30–90 day window is needed to see full recovery. · Customer Response Rate is modelled via demand elasticity assumption, not observed behavioural data.